B.Tech. ELECTRONICS AND COMMUNICATION ENGINEERING

COURSE PLAN: THEORY COURSE

Department :	**Electronics and Communication Engineering**			
Course Name & code :	**Signals and Systems & ECE2123**		**Core**	
Semester & branch :	**III & ECE**			
Name of the faculty :	**AV, PMP, DB, AKT, UV**			
No of contact hours/week:	**L**	**T**	**P**	**C**
	3	**0**	**0**	**3**

Course Outcomes (COs) to PO, PSO, BL Mapping

At the end of this course, the student should be able to:		**No. of Contact Hours**	**Marks**	**Program Outcomes (POs)**	**PSOs**	**BL *(Recommended)***
CO1	Analyse the time domain and transform domain characteristics of continuous-time periodic and Aperiodic Signals.	7	22	PO1, PO2, PO12	PSO2	2, 3, 4
CO2	Analyse continuous time LTI systems in the time domain and transform domain	8	22	PO1, PO2, PO12	PSO2	2, 3, 4
CO3	Analyse the time domain and transform domain characteristics of discrete-time periodic and Aperiodic Signals.	7	22	PO1, PO2, PO12	PSO2	2, 3, 4
CO4	Analyse discrete time LTI systems in the time domain and transform domain	10	26	PO1, PO2, PO12	PSO2	2, 3, 4
CO5	Understand and apply sampling theorem and its implications	4	8	PO1, PO2, PO12	PSO2	2, 3, 4
	Total	**36**	**100**			

Course Articulation Matrix

CO	Engineering knowledge	Problem analysis	Design/development of solutions	Investigations of complex problems	Modern tool usage	Engineer and society	Environment and sustainability	Ethics	Individual and team work	Communication	Project management and finance	Life-long learning				
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3										3			1	
CO2	3	3										3			1	
CO3	3	3										3			1	
CO4	3	3										3			1	
CO5	3	2										3			1	
Average Articulation Level	3	2.8										3			1	

ICT Tools used in delivery and assessment

Sl. No	Name of the ICT tool used	Details of how it is used
1	Power point presentation	For course delivery
2	Light house platform	Assignment conduction evaluation, and upload course materials
3	Exam cloud	Conduction and evaluation of midterm
4	Impartus	Class recordings can be accessed by the students

Typical tools including LMS, Smart Boards, MS Teams, etc

Mapping of Course Outcomes (COs)/Course Learning Outcomes (CLOs) #

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes (POs)	Learning Outcomes (LOs)	BL *(Recommended)*
CLO1	Analyse the time domain and transform domain characteristics of continuous-time periodic and Aperiodic Signals.	8	22	PO2, PO4	**C1, C2, C3, C18**	2, 3, 4
CLO2	Analyse continuous time LTI systems in the time domain and transform domain	8	22	PO2, PO4	**C1, C2, C3, C6, C18**	2, 3, 4
CLO3	Analyse the time domain and transform domain characteristics of discrete-time periodic and Aperiodic Signals.	6	22	PO2, PO4	**C1, C2, C3, C18**	2, 3, 4
CLO4	Analyse discrete time LTI systems in the time domain and transform domain	10	26	PO2, PO4	**C1, C2, C3, C6, C18**	2, 3, 4
CLO5	Understand and apply sampling theorem and its implications	4	8	PO2, PO4	**C1, C2, C3, C18**	2, 3, 4
	Total	**36**	**100**			

Applicable to IET Accredited Courses (modules) Only

Delivery and Assessment Plan of LOs #

Learning Outcome (LO) mapped to the course		Delivery and assessment Plan
LO	**LO statement**	
C1	Apply knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Some of the knowledge will be at the forefront of the particular subject of study.	Classroom teaching/ assignment, quizzes, test and end semester exam
C2	Analyse complex problems to reach substantiated conclusions using first principles of mathematics, statistics, natural science and engineering principles	Classroom teaching/ assignment, quizzes, test and end semester exam
C3	Select and apply appropriate computational and analytical techniques to model complex problems, recognising the limitations of the techniques employed	Classroom teaching/ assignment, quizzes, test and end semester exam
C6	Apply an integrated or systems approach to the solution of complex problems	Classroom teaching/ assignment, quizzes, test and end semester exam
C18	Plan and record self-learning and development as the foundation for lifelong learning/CPD	Classroom teaching

Applicable to IET Accredited Programs Only

Assessment Plan (As communicated from o/o AD-A, in every odd semester)

IN – SEMESTER ASSESSMENTS									
Sl. No.	**Assessment Mode**		**Assessment Method**	****Time Duration**	****Marks**	**** Weightage**	**Typology of Questions *(Recommended)***	**Schedule**	****Topics Covered**
1	**MISAC**	**1**	**Assignment**	**15 Minutes**	**05**	Descriptive	Bloom's taxonomy (B) level of the question should be L3 and above.	11-19 ,Aug 2025	L1-L7
		2	**Mid-Term Test**	**90 Minutes**	**30**	10 MCQs × ½ = 5 Descriptive= 25	Bloom's taxonomy (BT) level of the question should be L3 and above.	15-18 Sep	L1-L17
		3	**Assignment**	**15 Minutes**	**05**	Descriptive	Bloom's taxonomy (BT) level of the question should be L3 and above.	1-8 Sep 2025	L8-L15
2	**FISAC**	**1**	**Quiz**	**--**	**10**	**10 Questions to be answered**	Bloom's taxonomy (BT) level of the question should be L3 and above.	13-21 Oct 2025 or 3-8,Nov 2025	
END – SEMESTER ASSESSMENT									
1	**Regular/Make–Up Exam**			----	50	Answer all 5 full questions of 10 marks each. Each question can have 3 parts of 2/3/4/5/6 marks.	Bloom's taxonomy (BT) level of the question should be L3 and above.		

Lesson Plan

L No	Topics	CO Addressed
0	*To address OBE philosophy towards NBA and IET accreditation followed by a discussion on Course Plan, Course Outcomes, CO-PO Mapping and Blooms Taxonomy, with prerequisites for the course, if any.*	
1	Introduction to Signals and Systems	1
2	Classification of Continuous Time(CT) signals	1
3	Classification of Dicrete Time (DT)signals	3
4	Elementary CT signals	1
5	Elementary DT signals	3
6	Operations on dependent variables and independent variables of CT signals	1
7	Operations on dependent variables and independent variables of DT signals	3
8	Properties of CT systems-Memory, linearity, causality	2
9	Properties of CT systems- invertibility, time invariance and stability	2
10	Properties of DT systems-Memory, linearity, causality	4
11	Properties of DT systems- invertibility, time invariance and stability	4
12	Introduction to CT-Linear time invariant (LTI) systems and linear convolution	2
13	Introduction to DT-Linear time invariant (LTI) systems and linear convolution	4
14	Properties of CT-LTI Systems,examples	2
15	Properties of DT-LTI Systems,examples	4
16	Block diagram representation of CT systems	2
17	Block diagram representation of DT systems	4
18	Fourier representation for four classes of signals, Fourier series (FS)	1
19	FS representation of CT-periodic signals,Properties	1
20	DT-FS representation of DT-periodic signals, Properties	3
21	Fourier transform(FT) representation of CT-nonperiodic signals, properties	1
22	FT of signals, Worked out examples	1
23	Discrete-time FT (DTFT) of nonperiodic DT signals, properties	3
24	DTFT of signals, Worked out examples	3
25	Application of Fourier representation to CT-LTI systems,	2
26	Application of Fourier representation to DT-LTI systems	4
27	Sampling of CT signals,Sampling theorem.	5
28	Sampling of CT signals-continued with examples	5
29	Reconstruction of CT signals from its samples, Aliasing with examples	5
30	Ideal Reconstruction,Nyquist Theorem	5
31	Introduction to Z-transform	2
32	Poles and zeroes, stability	2
33	Region of convergence(ROC) ,Properties of ROC	4
34	Properties of Z-transform	4
35	Z-transform of DT functions	4
36	Transform Analysis of DT-LTI systems	4

Faculty members teaching the course (if multiple sections exist):

Faculty	Section
AKT	A
AV	B
PMP	C
DB	D
UV	VLSI

References:

Textbooks	• Simon Haykin, Barry Van Veen, "Signals and Systems", John Wiley &Sons, New Delhi, 2008. • V. Oppenheim, A. S. Willsky, A. Nawab, "Signals and Systems", PHI. Pearson Education, New Delhi, 2015. • H. Hsu, R. Ranjan "Signals and Systems", Schaums outline, Tata McGraw Hill, New Delhi, 2006. • P. Lathi, "Linear systems and Signals", Oxford University Press, 2005.
Self-Directed Learning	Generation of signals and Fourier analysis MATLAB toolbox
Research Literature/ Case Studies	• •
NPTEL/Coursera/any MOOC-based material	• •

Submitted by: Aparna V

(Signature of the faculty)

Date: 19-07-2025

Approved by: Dr Pallavi R Mane

(Signature of HoD)